Bahraini Currency Recognition with Deep Learning (Guided Lab)

- Author: Ali H. El-Kassas | Senior Data Scientist
- Created: Aug 2026

Lab Objectives

By the end of this lab, you will be able to:

- Collect and prepare real-world image data
- Build a **currency classification model**
- Evaluate model performance using proper metrics
- Understand limitations of classification

Big Picture

We will build an AI system that understands **Bahraini currency (BHD)**.

We will do this in **two stages**:

1. **Image Classification**
   → "What currency note is in this image?"

This mirrors how real-world computer vision systems are developed:
start simple → then go production-grade.

Target Classes (Example)

You may choose **at least 10 classes**, such as:

- 0.5 BHD
- 1 BHD
- 5 BHD
- 10 BHD
- 20 BHD

Classification:

Data Collection

You must collect images of Bahraini currency using:

- Mobile phones
- Different lighting conditions
- Different backgrounds (table, hand, wallet, desk)
- Different angles and distances

Minimum Dataset Size

Class	Minimum Images
Each currency	50 images
Total	500-1000 images

Tip: More data = better results

Preprocessing

Before training:

- Resize images (e.g. `224 × 224`)
- Normalize pixel values
- Apply **data augmentation**:
    - Rotation
    - Zoom
    - Horizontal flip
    - Brightness changes

Why augmentation?

Model Training (Classification)

Model Evaluation

Evaluate on the **test set** using:

- Accuracy

- Confusion Matrix
- Precision / Recall
- Per-class performance
- Error analysis

Model Deployment (streamlit)

Bonus: you can use FastAPI + FastAPI cloud

Key Questions

- Which currency is most confusing?
- Are similar-looking notes misclassified?
- Does background affect performance?

Real-World Testing

Test your model on:

- Live camera (streamlit app)
- Mobile phone photos

- New unseen backgrounds

Discuss:

- False positives
- Missed detections
- Speed vs accuracy trade-offs

Final Reflection

Students should answer:

1. Why did classification work well initially?
2. What problems did object detection solve?
3. Which approach is more suitable for real-world deployment?
4. How would you improve the system further?

Bonus Challenges (Optional)

- Detect **front vs back** of currency
- Count total money in an image